Trainings

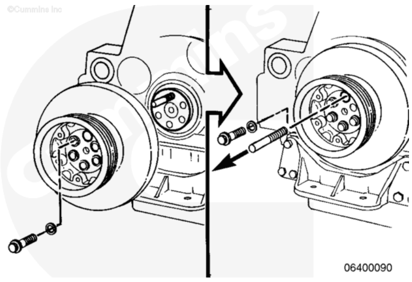
Preparatory Steps

- Remove the fan belt. Refer to Procedure 008-002 in Section 8 (/qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html).
- Remove the crankshaft pulley, if equipped. Refer to Procedure 001-022 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-022-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not pry or hammer on the vibration damper. Damage will result.

Remove one capscrew, and install an M14 X 105 mm guide stud.

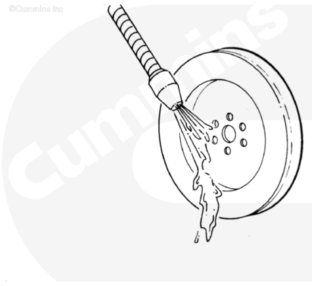
Remove the remaining capscrews and dampers.

Remove the guide stud.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



da100ec

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : Vibration dampers have a limited service life. The damper must be replaced after 24,000 hours in service in all applications except haul trucks. Haul truck dampers can be replaced at rebuild.

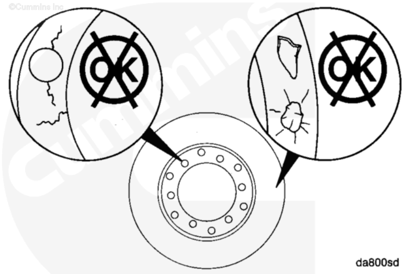
Do **not** repair or balance a viscous damper in the field.

Use solvent to clean the damper and dry with compressed air.

Inspect the vibration damper mounting flange for cracks.

Inspect the vibration damper housing for dents, bulges, or leaks.

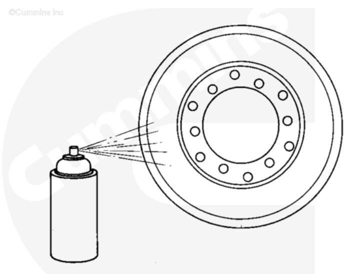
Replace the vibration damper if damaged.



da800sd

Use Crack Detection Kit, Part Number 3375434, to check the vibration damper for cracks.

If the vibration damper is cracked, it **must** be replaced.

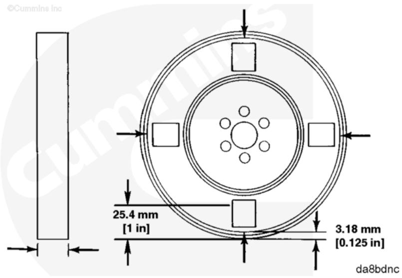


da800se

Measure

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use a paint solvent and fine emery cloth to remove paint from the front and back of the housing at the four locations shown in the illustration.

Damper thickness measurements are to be taken **no** less than 3.0 mm [0.125 in] and 25.4 mm [1.0 in] from the outside circumference.

Measure the thickness at two points in four locations around the damper, 90 degrees apart.

Maximum Vibration Damper Thickness		
mm		in
65.66	MAX	2.585

The readings **must not** vary more than 0.25 mm [0.010 in] from one another.

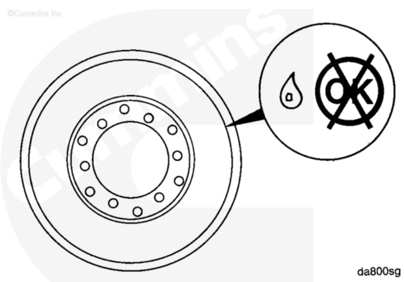
If the vibration damper is **not** within specifications, the vibration damper **must** be replaced.

If the vibration damper has been in service for 24,000 hours or more, the vibration damper **must** be replaced regardless of the thickness measurement. Haul truck dampers can be replaced at rebuild.

Leak Test

⚠ CAUTION ⚠

Dampers that have exceeded their useful life will not function properly and can cause engine damage because of excessive torsional vibration in the engine.



Remove the vibration damper.

Check for fluid leakage around the rolled lip.

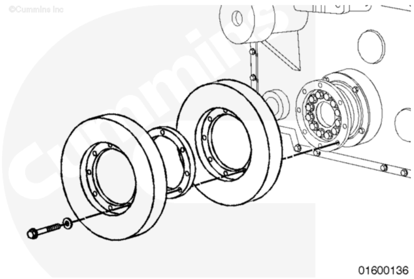
The vibration damper **must** be replaced if there is any fluid leakage.

Note : Always replace the vibration damper after a crankshaft nose malfunction. Crankshaft nose malfunctions are primarily caused by excessive torsional activity.

Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



The vibration damper with the timing marks goes on first.

Use a M14 X10.5 mm guide stud to aid in installation.

Install the vibration damper on the crankshaft adapter.

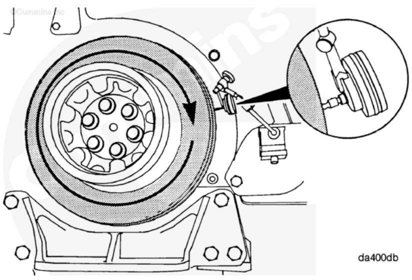
Install the eight capscrews.

Tighten the capscrews.

Torque Value:

1. 125 n•m [92 ft-lb]
2. 165 n•m [122 ft-lb]

Eccentricity Check



Use a dial indicator and adjust it, as shown, to measure radial alignment of the vibration damper. Turn the crankshaft 360 degrees. Record the total indicator runout.

Both vibration dampers **must** be checked.

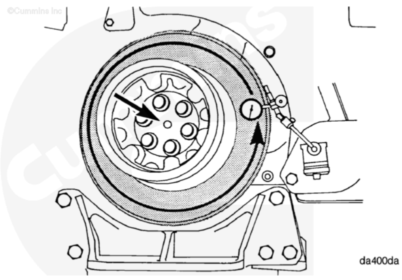
Vibration Damper Eccentricity		
mm		in
0.46	MAX	0.018

If the measurement is **not** within specifications, make sure the vibration damper pilot is aligned properly.

Wobble Check

The crankshaft end clearance **must** be pushed or pulled in the same direction each time a measurement is taken. Begin with the crankshaft in the full front or rear position.

Vibration Damper Face Runout		
mm		in
0.56	MAX	0.022



The indicator contact point **must** be placed 13.0 mm [0.50 in] inward from the outer edge of the vibration damper.

If the measurement is **not** within specifications, check for foreign material between the crankshaft and the adapter (or pulley), and between the adapter (or pulley) and the vibration damper.

Finishing Steps

- If equipped, install the crankshaft pulley. Refer to Procedure 001-022 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-022-tr.html)
- Install the fan belt. Refer to Procedure 008-002 in Section 8 (/qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html).

Last Modified: 14-Oct-2019